

Researchers from Cambridge and collaborators from the US and Spain are investigating three-dimensional nano-fabrication of silicon chips using electron microscopes, which are one of the most versatile tools in nanotechnology. The purpose of this investigation is to bring a radical change by increasing the storage capacity of silicon chips through 3D fabrication. The usual chips are fabricated in a plane, that is, these are two dimensional and hence have less storage capacity.

These researchers from Cambridge have tried to combine electron microscopic processes (electron emissions) with gases in order to use these microscopes as 3D nano printers for the purpose of 3D fabrication. The same group is investigating the properties of nanomagnetic wires for computations, but they are facing a couple of challenges. First, the materials needed must be of good quality for better data flow. Second, 2D nanostructures have limited memory storage capacity. Still, researches and works like these are undoubtedly the stepping stones in the field of green computing and need major attention.

In India, the Ministry of Electronics and Information Technology (MeitY) has sponsored mega facility on nanoelectronics led by IIT Bombay and IISc, called the Centre of Excellence in Nano-electronics (CEN). This facility is working on nanofabrication and development of devices for low power consumption, energy-saving, nano-biosensing, and communication in the domain of healthcare, communication, agriculture, and IT.

More than 37 projects have been completed as of now. A few more projects have been sponsored by DST, DRDO, and other agencies in India on green IT and electronics.

MeitY has also sponsored the Indian Nanoelectronics User Programme (INUP) and Nanoelectronics Network for Research and Applications

(NNetRA) to support various activities of green IT and electronics in India.

India has shown quite a positive response towards green IT and electronics. It has already become one of the major concerns for investors, manufacturers, policymakers, and other stakeholders to adopt green IT methodologies and upgrade the existing infrastructure in related companies across India and make it energy-efficient. The achievements of the Indian government, with respect to green IT and computing in India, are as follows:

1. The government of India included green IT in 12th Five-Year Plan with a few guidelines for IT and telecom industries in India such as:
 - (a) Standardisation for electronic goods and products with energy consumption as a technical standard.
 - (b) Setting up a centre of excellence in top technological institutes and universities, which is responsible for developing and certifying the prototypes and end products.
 - (c) Tax incentives to those who adopt greener technologies.
 - (d) All the carbon footprints released in network-related operations would have to be declared to TRAI.
 - (e) Service providers were supposed to aim their carbon emission reduction targets for the mobile network at 17.5 per cent by the year 2018-19.
2. Global ICT Standardisation Forum for India was introduced in order to focus on the development of ICT

(Information and Communication Technology) and bridge Indian achievements globally.

3. The smart meters deployment initiation by Tata Power Delhi Distribution Limited and BSES substantially reduced revenue loss incurred through AT&C losses.
4. Advanced metering infrastructure, distribution automation, and distributed energy resources, which form an integrated set up of ICT, yield a lot of benefits with respect to power consumption.
5. AI induced ICT tools that add on to green IT applications (like sensors, smart meters, GIS, and SCADA) enable quick identification, prediction, and prevention of expected problems such as air pollution, burst water leaks, non-transparent billing, contamination checks, etc.

Conclusion

It is quite exciting that we can get really close to green computing and electronics with devices in our hands, which are based on greener technologies, processors, and memories designed with smart architectures. Green IT would make these devices less dependent on energy and hence help save it for the betterment of environmental sustainability.

Moreover, this would not only reduce the amount of e-waste but also help save natural resources like water, which is used as a coolant in data centres. However, there is still a long way to go, and we all need to be conscious about energy and environmental concerns as well. **END** 

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The article was originally published in the September 2020 issue of Electronics For You.